

Researcher Profile

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- Google Scholar: [7TCwAaUAAAAJ](https://scholar.google.com/citations?user=7TCwAaUAAAAJ)

Research Interests

- Topological Data Analysis (TDA) and topological signal processing.
- Dynamical systems and bifurcation/chaos detection.
- Machine learning and deep learning for time-series and signals.
- High-performance computing and GPU-accelerated AI (multi-node/multi-GPU).
- Large language models and scalable local inference.

Education

- 2018–2023 Ph.D. in Mathematics Engineering
Istanbul Technical University
Thesis: Topological Data Analysis and Clustering Algorithms in Machine Learning
Advisor: [Dr. Atabey Kaygun](#)
Project: The Scientific Research Projects İTÜ: TDK-2020-42698
- 2014–2016 M.Sc. in Mathematics
Dokuz Eylül University GPA: 3.88/4.00
Thesis: On Highest Eigenvalues of Regular Sturm–Liouville Problems
Advisor: [Dr. Şennur Somali](#)
- 2008–2014 B.Sc. (Major) in Mathematics
Dokuz Eylül University GPA: 3.81/4.00
magna cum laude
- 2014–2015 Pedagogical Formation in Mathematics
Dokuz Eylül University
- 2012–2015 B.Sc. (Minor) in Statistics
Dokuz Eylül University GPA: 3.29/4.00
- 2012–2013 Exchange Program, Aarhus University, Denmark GPA: 3.66/4.00
Project: Solving Equations via Eigenvalues and Eigenvectors
Advisor: [Dr. Anders Nedergaard Jensen](#)

Academic and Research Appointments

- 2023–present **Senior Researcher-Project Researcher**, TRUBA HPC Centre, Network Technologies Department, National Academic Network and Information Center, [TÜBİTAK ULAKBİM](#), The Scientific and Technological Research Council of Türkiye
- 2024–present **Part-time (EK-46) Researcher and Lecturer, Actuarial Sciences**, Institute of Applied Mathematics, [METU](#)
 - Teaching: [IAM788 – Topological Machine Learning in Time Series](#).
 - Co-supervisor of Ph.D. student.
 - Researcher on TÜBİTAK ARDEB 1001 Project(124F095): Detection of Stationarity in Time Series with Artificial Intelligence.
- 2021–2022 **Visiting Researcher (12 months during Ph.D)**, Michigan State University (via TÜBİTAK 2214-A)
Advisor: [Dr. Elizabeth Munch](#)
Project: Topological Data Analysis and Machine Learning Algorithms for Real-World Problems
- 2018–2023 **Research and Teaching Assistant**, Istanbul Technical University
 - Assisted courses: Mathematics I, Mathematics II, Introduction to Programming (C).
 - Responsibilities included running tutorials/labs, preparing materials, grading, and student support.

Industry and Other Experience

- 2019–2022 **Quantitative Analyst-Part Time**, [Algotecht](#)
2020–2022 **Lecturer, Machine and Deep Learning Courses-Part Time**, [UcuncuBinYil](#)
2015–2018 **Mathematics Teacher**, BAYSEM, Bayrakli, Izmir

Publications

Journal Papers — Mathematics

- 2026 *Persistence Landscapes for Topological Analog Retrieval in Market Risk Forecasting*, 2026 (submitted to [Chaos, Solitons and Fractals](#); under review).
Güzel, İ.
- 2026 *Persistent Homology of Time Series through Complex Networks*, [arXiv preprint arXiv:2605.01624](#), 2026.
Güzel, İ.
- 2026 *Robust Short-Term OEE Forecasting in Industry 4.0 via Topological Data Analysis*, [arXiv preprint arXiv:2507.02890](#), 2026 (under review).
Anapa, K., Güzel, İ., Yozgatligil, C.
- 2025 *Teaspoon: A Python Package for Topological Signal Processing*, [Journal of Open Source Software](#), 2025.
Khasawneh, F. A., Munch, E., Barnes, D., Chumley, M. M., Güzel, İ., Myers, A. D., ... & Yesilli, M.
- 2023 *Classification of Stochastic Processes with Topological Data Analysis*, [Concurrency and Computation: Practice and Experience](#), 2023.
Güzel, İ., Kaygun, A.
- 2022 *Detecting Bifurcations in Dynamical Systems with CROCKER Plots*, [Chaos: An Interdisciplinary Journal of Nonlinear Science](#), 2022. **Featured Article**.
Güzel, İ., Munch, E., Khasawneh, F.
- 2022 *Persistent Homology, Matroids and Cobordisms*, [arXiv](#), 2022.
Güzel, İ., Kaygun, A.
- 2022 *A New Non-Archimedean Metric on Persistent Homology*, [Computational Statistics](#), 2022.
Güzel, İ., Kaygun, A.
- 2019 *Operator Splitting Methods for Computation of Eigenvalues of Regular Sturm–Liouville Problems*, [Surveys in Mathematics and its Applications](#), 2019.
Güzel, İ., Adıyaman, M., Somali, S.

Journal Papers — Interdisciplinary

- 2024 *Bacterial Microbiome Diversity Along Poultry Slaughtering Lines: Insights from Chicken Carcasses and Environmental Sources*, [Journal of Veterinary Research](#), 2024.
Telli, A. E., Biçer, Y., Telli, N., Sönmez, G., Turkal, G., Güzel, İ.
- 2023 *Phenolic Constituents, Antioxidant and Antimicrobial Activity and Clustering Analysis of Propolis Samples Based on PCA from Different Regions of Anatolia*, [Molecules](#), 2023.
Altuntaş, Ü., Güzel, İ., Özçelik, B.
- 2022 *Attitudes and Behaviors of Turkish Consumers Regarding the Olive Oil Consumption*, [Erzincan University Journal of Science and Technology](#), 2022.
Altuntaş, Ü., Güzel, İ., Yılmaz, Ö., Uluata, S., Özçelik, B.

Conference Papers

- 2026 *BeTiSe: A Comprehensive Benchmark Dataset for Time Series Stationarity and Structural Analysis*, [ITISE 2026](#) (12th International Conference on Time Series and Forecasting), accepted.
Gür, K. R., Yazıcı, P. C., Erkaya, P., Türkmen, Y., Baytak, B., Güzel, İ., Karagöz, P., Yozgatligil, C.
- 2024 *An Overall Equipment Efficiency Predictive Analysis of a Hydraulic Press System by Time Series Forecasting with Topological Features*, [\(IDSSC 2024\) International Data Science and Statistics Congress: Artificial Intelligence](#), October 2024.
Anapa, K., Güzel, İ., Yozgatligil, C.
- 2022 *Classification of Stochastic Processes with Topological Data Analysis*, [BAŞARIM 2022 – 7th High-Performance Computing Conference](#), May 2022.
Güzel, İ., Kaygun, A.
- 2022 *A Case Study on Identifying Bifurcation and Chaos with CROCKER Plots*, [Proceedings of TDA at SDM \(SIAM Data Mining\)](#), April 2022.
Güzel, İ., Munch, E., Khasawneh, F.

Submitted / Under Review

- 2026 *Towards Sustainable Text-to-SQL Inference on HPC Infrastructure*, [NeurIPS 2026 Evaluations and Datasets Track](#), submitted.
Erdogan, M., Erfidan, M., Güzel, İ., Toroslu, I. H., Karagöz, P.
- 2026 *Enhancing First-Shot Anomalous Sound Detection in Noisy Industrial Environments*, [IEEE 28th International Workshop on Multimedia Signal Processing \(MMSP\)](#), submitted.
Anapa, K., Güzel, İ., Yozgatlıgil, C.

Posters

- 2021 *Bifurcation Analysis with CROCKER Plots*, [CMSE Student Research Symposium at Michigan State University](#), Nov 2021.
Güzel, İ., Munch, E., Khasawneh, F.
- 2021 *Hierarchical Clustering and Zeroth Persistent Homology*, [Institute for Mathematical and Statistical Innovation \(IMSI\)](#), May 2021.
Güzel, İ., Kaygun, A.

Academic Activities

Seminar and Conference Talks

- 2024 Artificial Intelligence and High Performance Computing Ecosystem, [BAŞARIM2024](#), May 2024.
- 2022 Classification of Stochastic Processes with Topological Data Analysis, [BAŞARIM2022](#), May 2022.
- 2022 A Case Study on Identifying Bifurcation and Chaos with CROCKER Plots, [TDA Workshop at SDM 2022](#).
- 2021 A Guided Tour of Data Science Ecosystem for Asset Managers, Quant Workshop, Chartered Financial Analyst (CFA) Istanbul Society, June 2021.
- 2016 *Computing Eigenvalues of Regular Sturm–Liouville Problems*, The 3rd Workshop of Association for Turkish Women in Maths, May 27–29, Izmir, Turkey (joint with Ş. Somalı).

Selected Workshops and Courses

- 2025 [Trillion Parameter Consortium 2025 \(TPC25\)](#), San Jose, California, US, July 28–31, 2025.
- 2025 [ISC High Performance 2025](#), Hamburg, Germany, June 10–13, 2025.
- 2024 Multi-GPU Deep Learning, [HLRS, University of Stuttgart](#), Stuttgart, Germany, July 02–05, 2024.
- 2023 Modern Artificial Intelligence and Mathematical Foundations, Bilimler Köyü, Foça, İzmir, July 24–30, 2023.
- 2023 ACM Europe Summer School on HPC Computer Architectures for AI and Dedicated Applications, Barcelona Supercomputing Center, Barcelona, Spain, July 02–07, 2023.
- 2022 Topological Data Visualization Workshop, University of Iowa, May 16–20, 2022.
- 2021 Computational Persistence (ComPer) Workshop, Purdue University, Nov 1–5, 2021.
- 2021 Summer School: Theory and Practice of Deep Learning, Gene Golub SIAM Summer School, July 19–30, 2021.
- 2021 Quantum Information for Mathematics, Economics, and Statistics, Institute for Mathematical and Statistical Innovation (IMSI), May 24–28, 2021.
- 2020 İMO2020 – Machine Learning in İstanbul (14 hours theoretical, 14 hours implementation), Mimar Sinan Fine Arts University, Feb 2020.

Teaching

- 2024–present [IAM788 – Topological Machine Learning in Time Series](#), Institute of Applied Mathematics, METU
- 2023–present PyTorch Lightning on Multi-GPU AI (Train-the-Trainers), [National Competence Centers Türkiye](#)
- 2023–present GPU Usage for AI and Feature Engineering, National Technology Initiative – AI Specialization Program
- 2023–present Introduction to HPC and Container-based Computing (Docker, Singularity/Apptainer), TÜBİTAK ULAKBİM
- 2020–2022 Machine Learning and Deep Learning Courses, [UcuncuBinYil](#)
- 2018–2023 Teaching Assistant: Mathematics I, Mathematics II, Introduction to Programming (C), İstanbul Technical University

Academic Service

- 2026 [Chaos: An Interdisciplinary Journal of Nonlinear Science](#) (AIP Publishing)
- 2025 [La Matematica](#)
- 2022 [Journal of Open Source Software \(JOSS\)](#)

- 2022 [Topology, Algebra, Geometry in Machine Learning \(TAG-ML\), ICML 2022](#)
2022 [Geometrical and Topological Representation Learning, ICLR 2022](#)
2021 [Geometrical and Topological Representation Learning, ICLR 2021](#)

Technical Skills

- CAS Mathematica, SageMath
Languages Python, R, Julia, Matlab, Paraview, Bash
Frameworks NVIDIA Containerized AI Tools, TensorFlow, PyTorch, XGBoost, Sktime; Accelerate, Horovod, DeepSpeed, Ray; MPI, OpenMP
Tools High-Performance Computing, Linux, $\LaTeX$, Inkscape, Docker, Singularity/Apptainer, Enroot Git

Professional Development

- Data Parallelism: How to Train Deep Learning Models on Multiple GPUs, NVIDIA
Applications of AI for Anomaly Detection, NVIDIA
Fundamentals of Accelerated Computing with CUDA Python, NVIDIA
Fundamentals of Deep Learning, NVIDIA
High Performance Data Analysis with R, IT4Innovations National Supercomputing Center

Fellowships & Grants

- 2026–2029 Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA-PF), European Commission (approx. €400,000 total budget).
Host: Middle East Technical University (METU); Partner: Michigan State University.
2024–present (Researcher) Detection of Stationarity in Time Series with Artificial Intelligence, 124F095, TÜBİTAK ARDEB 1001
2021–2022 (PI) Topological Data Analysis and Machine Learning Algorithms for Real-World Problems, TÜBİTAK 2214-A
2020–2024 (Researcher) Topological Data Analysis and Clustering Algorithms on Machine Learning, TDK-2020-42698, Istanbul Technical University

Scholarships

- 2017–2021 The Scientific and Technological Research Council of Türkiye, Grant 2211-A
2014–2016 Republic of Türkiye, State Scholarship (for masters education)
2011–2014 Zorlu Foundation Fellowship

Honors & Awards

- 2014 Graduated first in Science Faculty, Dokuz Eylül University
2011–2014 Certificate of High Honor by Science Faculty Dean, Dokuz Eylül University

Languages

- Turkish (Native), English (Professional)

References

- [Dr. Atabey Kaygun – *kaygun@itu.edu.tr*](#)
- [Dr. Elizabeth Munch – *muncheli@msu.edu*](#)
- [Dr. Selçuk Bilmiş – *selcukbilmis@gmail.com*](#)
- [Dr. Ceylan Yozgatlıgil – *ceylan@metu.edu.tr*](#)